RITIK RAMESHWAR PATEL

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Available full-time • immediate start • F-1 • CPT/OPT eligible

WORK EXPERIENCE

Research Assistant | Arizona State University – Speech/Language Genetics Lab – Tempe, AZ | May 2026 – Present

- Building a React Native dyslexia support application with structured logging for speech, task accuracy and adaptive difficulty; session data lands in PostgreSQL (AWS RDS) via Python ETL so cognitive performance is queryable for analysis.
- Prototyping ML and analytics for the lab: user clustering, difficulty recommendation, task performance prediction and a researcher dashboard to track progress and session quality on early app data.

Graduate Teaching Assistant | Arizona State University – W. P. Carey School of Business – Tempe, AZ | Jan 2025 – May 2026

Courses: Analytics for Unstructured Data; ML in Business; Programming for AI and Data Analytics in Business

- TA for 3 graduate courses (~150 students per course, 20 hrs/week). Taught and graded weekly assignments, quizzes and semester projects on data analytics, statistical analysis, ML/AI, LLM integration, prompt evaluation, vector databases and NLP (sentiment analysis, BERTopic, multi-LLM tasks).

Data Analyst | WeAgile Software Solutions – Pune, India | Mar 2022 – Jun 2024

Clients: Meta, Microsoft, The Home Depot, Ralph Lauren, Wayfair, Waste Management

- Owned daily KPI reporting and business intelligence deliverables from CCTV analytics across 20+ cameras/day (~18–20 hrs of footage) and high-volume event data (footfall, dwell, queues, product touches, self-checkout flags, shrink alerts, bin fill) – data visualization in Power BI, Tableau and Excel.
- Built a real-time Power BI dashboard and a Python + Flask + ETL pipeline (MySQL) that replaced manual pulls from four different dashboards. Report prep time went from hours to minutes (60%+ faster).
- Designed data modeling for client dashboards – joins across event, camera and site tables, views for reusable KPI logic and indexes on high-traffic filters. Added data quality via validation queries (row counts, date gaps, duplicates) and triggers to block bad writes. Used window functions for ranked trends and rolling daily metrics before Power BI.
- Boosted the model-driven event quality by 20%+ across accounts and 38% on Home Depot self-checkout system (OpenCV, TensorFlow, Scikit-learn, PyTorch) – root cause analysis on false alerts (label noise, preprocessing, threshold drift) improved loss-prevention follow-up.
- Coordinated Android and iOS app delivery as project lead – Jira for tasks and releases, cross-team alignment and assets on AWS S3 and SharePoint.

Trainee Software Engineer | WeAgile Software Solutions – Pune, India | Nov 2021 – Feb 2022

- Labeled ~500 images/day in CVAT (bounding boxes, class tags, QA rework) for object detection datasets – mostly Waste Management, Wayfair warehouses and Irvine Spectrum Center – so models and downstream event counts were usable.
- Built ~2 floor plan 3D layouts/day in IP Video System Design Tool for Meta and Microsoft promo events and mall sites globally – camera placement to improve coverage before footfall and interaction tracking went live.

SKILLS

- **Analytics and Programming:** Python, SQL, PostgreSQL, MySQL, Pandas, NumPy, Flask, Git
- **Data and Cloud:** Power BI (DAX), Tableau, Advanced Excel, AWS (S3, Athena), Docker, SharePoint, Jira
- **Analysis and ML:** Scikit-learn, TensorFlow, PyTorch, OpenCV, Predictive Analytics, Data Validation, NLP, LLMs

EDUCATION

Master of Science in Data Science, Analytics and Engineering | May 2026

Arizona State University – Tempe, AZ

Relevant Coursework: Data Analytics, Data Mining, Database Management, Data Visualization, Statistical Machine Learning, Optimization, NLP

Master of Computer Applications (MCA) | May 2022

Savitribai Phule Pune University – Pune, India

Relevant Coursework: Database Management Systems, Data Analytics, Machine Learning, Software Engineering

RESEARCH AND PROJECTS

Wildfire Detection with Social Media Signals | Jan 2026 – May 2026

- Multimodal capstone merging satellite fire imagery and Reddit posts: ResNet-18 on multi-fire image sets with overlapping patches, plus a BERT text classifier and spaCy NER for place names from unstructured posts.
- Built separate train/validation splits, error analysis and reporting on precision, recall, F1 and ROC-AUC for vision and NLP tracks; goal is earlier corroboration and geographic context when camera coverage or cloud cover is limited.
- Built ETL pipelines to ingest wildfire imagery and Reddit text into a shared event schema with geospatial fields; ran data quality checks (missing timestamps, duplicate posts, empty image patches) before scores fed the multimodal alert layer.

ML Paper on Network Intrusion Detection (NIDS) | Aug 2024 – Dec 2024

- End-to-end ML paper on NSL-KDD and CIC-IDS binary and multi-class attack labels, preprocessing, feature selection and model benchmarking in Python (scikit-learn, XGBoost).
- Achieved ~99% classification accuracy on held-out NSL-KDD test data across attack classes; ran head-to-head tests vs decision trees, gradient boosting and XGBoost, scoring each on false alarms vs catch rate, precision/recall and stability on imbalanced labels for near-real-time IDS use; documented full methodology, evaluation metrics and model settings so the experiments could be rerun and reported consistently in the paper.

Retail KPI Analytics Pipeline | Aug 2025 – Dec 2025

- End-to-end analytics pipeline modeled on retail/venue CCTV KPI work: ~90 days of site/camera/event logs (10+ sites, 50+ cameras, 1k+ events/day), AWS S3 (partitioned by date), Athena SQL, PostgreSQL data modeling (staging/mart tables, 15+ queries/views), Power BI with 8+ DAX measures (daily footfall, 7-day rolling trends, site rankings, Week-over-Week %).
- Worked with Docker on Python ingest/refresh job (connects Athena to Postgres); GitHub repo with architecture diagram, sample SQL and refresh docs – built to replace a manual multi-dashboard copy-paste workflow similar to prior reporting automation work.
- Implemented data quality checks before each refresh (row counts, duplicate event IDs, null-rate monitoring and date-gap detection), improving dashboard reliability and preventing KPI drift across reporting periods.